Spencer Presley

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EXPERIENCE

CrunchAtlas — Software Engineer | Nov 2025 – Present

Local-first, air-gapped AI security platform for critical infrastructure. Sole author of the systems below.

- **Replaced Celery with a Postgres-native job queue** for the platform's GPU, LLM, sandbox, and document jobs, so any one job can be cancelled without killing its worker. Interactive work runs ahead of background work, customers take fair-share turns, dependency DAGs gate what is claimable, and workers claim with `SELECT FOR UPDATE SKIP LOCKED`.
- **Made every job reach a terminal state, even when its process dies.** Each runs under a spawned (not forked) supervisor that survives a segfault or OOM kill and reaps the child's process group, and a heartbeat sweeper fails jobs whose supervisor dies or freezes. Chaos tests SIGKILL live supervisors and kill the database mid-job.
- **Replaced fixed LLM concurrency with token-priced GPU admission.** Five fixed 131k-token llama.cpp slots became one 655k-token KV pool that holds 7 concurrent 128k-context agents or 160 4k-token jobs; a per-pool Postgres advisory lock admits a job only when its tokens fit, re-pricing under `READ COMMITTED` after any wait.
- **Made the platform investigate on its own**, completing nearly 17,000 PCAP investigations in its first two months with no analyst click. A background reconciler queues the AI work that should exist; each dispatch commits independently behind a durable resume cursor.
- **Cut the Files page query from 12.3 s to 1.5 ms** on a 7.2M-row table by moving per-document alert counts into a summary table kept exact by statement-level Postgres triggers, with a reconcile job that repairs drift.
- **Created the agent harness every security agent runs on** (LangChain/LangGraph). Before each model call it tokenizes the exact outgoing request and compacts or refuses anything that won't fit; it also evicts oversized tool output, detects tool-call loops, and sandboxes each run in a network-isolated container.
- **Own AtlasCyber's AWS across GovCloud and commercial partitions**, managed in Terraform: EC2 and its reverse-proxy/tunnel fleet, load balancers, WAF, S3, IAM, VPC networking, CloudWatch alarms, and CloudFront.
- **Rewrote the network-capture sensor from scratch in Rust** (`#![forbid(unsafe_code)]`): a headless Linux daemon whose SQLite-WAL upload queue persists each capture before upload, so queued captures survive a restart.

CrunchAtlas — Software Engineer, AtlasConnect | Mar 2025 – Nov 2025

Sole developer and maintainer of an inherited codebase — Django REST, React/Vite, self-hosted LLM stack.

- **Turned pitch decks into structured deal fields** with a document-ingestion pipeline: PDF/DOCX/PPTX/DocSend extraction, parallel OCR only on pages without a text layer, and concurrent LangChain chains under a token budget.

PROJECTS

docq — *Python, SQLite/FTS5, Ollama*

- Created a cited retrieval engine that lets coding agents search a long reference without loading it. On a 266-case eval, hybrid BM25/vector retrieval beat either channel alone at 0.67 hit@1; a query-rewrite A/B raised it to 0.74 (p=0.0045).

Celery Fork-Safety Investigation — *Python, Celery, GDB, LLDB, Docker*

- Traced a Celery hang to fork safety: a Rust-backed vector store initialized before `fork()` left the child holding a lock whose owning thread didn't survive the fork. Linux deadlocked on a futex (GDB); macOS crashed in libdispatch (LLDB).

HONORS

- **1st Place / Best Hack — Bitcamp 2024**, UMD (largest US East Coast hackathon) — TermBook
- **Best Educational Hack & 2nd Overall — HackUMBC 2024** — Testif.AI

SKILLS

- **Backend & infrastructure:** Django, FastAPI, PostgreSQL, SQLite, Redis, Celery, Linux, AWS, Terraform
- **DevOps & CI/CD:** Docker (multi-stage, BuildKit), GitHub Actions, Jenkins, supervisord, systemd
- **Languages:** Python, Rust, TypeScript, SQL, Bash
- **AI & LLM:** LangGraph/LangChain, llama.cpp, llama-swap, Ollama, LLM serving, GPU capacity planning

EDUCATION

Salisbury University — B.S. Computer Science, Minor in Mathematics | Aug 2021 – May 2025